MILITARY
SPECIFICATION

MIL-S-12479B(SL)
20 November 1963
Superseding
MIL-S-12479A(SigC)
3 July 1957

SWITCHBOARD, TELEPHONE, MANUAL SB-993( )/GT (UNITS OF)

1. SCOPE

1.1 This specification covers the following units of one type of switchboard equipment designated Switchboard, Telephone, Manual SB-993( )/GT: (See 6.1 and 6.3)

Adapter Plug U-184( )/GT
Case, Switchboard CY-2589( )/GT.
Holder, Adapter, Connector MT-2156( )/GT

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-591	Boxes, Fiberboard, Wood-Cleated.
PPP-B-601	Boxes, Wood, Cleated Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Fiber.
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-P-116	Preservation, Methods of.
MIL-B-10377	Boxes, Wood, Cleated, Veneer, Paper-Overlaid.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attribute.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-169	Extreme-Temperature Cycle.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal and Electronic Test Equipment.

FSC 5805

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MILITARY (Cont'd)

MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects For.

DRAWINGS

ELECTRONICS COMMAND

SC-DL-145544	Holder, Adapter, Connector MT-2156( )/GT
SC-DL-145576	Case, Switchboard CY-2589( )/GT
SC-DL-145577	Adapter, Connector U-184( )/GT
SC-A-362100	Preparation of Preproduction Packs.

(Copies of specifications, standards, and drawings, required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- Switchboard, Telephone, Manual SB-993( )/GT is a small portable local battery, field type telephone switchboard. It comprises a Case, Switchboard CY-2589( )/GT, a Holder, Adapter, Connector MT-2156( )/GT and seven Adapter, Connector U-184( )/GT. The switchboard accomodatessix lines and one operator's telephone set. The glow lamp in each adapter signals an incoming call.

3.2 Preproduction samples.- The contractor shall furnish two preproduction samples of Switchboard, Telephone, Manual SB-993( )/GT for approval, if required by the Invitation for Bids and the Contract. (See 4.3 and 6.2)

3.3 Construction.- Switchboard, Telephone, Manual SB-993( )/GT shall be constructed in accordance with the drawings listed on the following Drawing and Data Lists:

Item	Drawing
Case, Switchboard CY-2589( )/GT	SC-DL-145576
Holder, Adapter, Connector MT-2156( )/GT	SC-DL-145544
Adapter, Connector U-184( )/GT	SC-DL-145577

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3.4 Finish.- The switchboard shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4)

3.5 Marking.- Marking shall conform to Specification MIL-M-13231. (See 4.4)

3.6 Tropicalization of material.-

3.6.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.6.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.6.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.6.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2)

3.7 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.9) Individual items shall not be hand-picked for fit or performance; [illegible], Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.8 Service conditions.- The equipment shall meet the following service conditions--where a test is referenced, meeting the test shall be considered as compliance with the requirement:

3.8.1 Operation.- Continuous use for a period of 10 years, under the worldwide environmental conditions specified in 3.8.2 through 3.8.5, with a duty cycle of 16 hours on and 8 hours off and with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement. Approval of the preproduction sample (see 3.2) shall be considered as compliance with this requirement.

3.8.2 Temperature (see 4.6.1).-

(a) Operating: Ambient temperature in the range of +132°F to -40°F. (The 132°F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

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(b) Nonoperating: Exposure in the range of +160°F to -60°F; exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.8.3 Relative humidity.- Exposure to 97 percent relative humidity for 20 hours and 100 percent relative humidity (with condensation) for 4 hours for each 24 hour period over the period of years stated in 3.8.1. Meeting the tests of 4.6.2 shall be considered compliance with this requirement.

3.8.4 Bounce.- The switchboard shall meet specified performance, with no physical damage allowed, after subjection to the test of 4.6.3.

3.8.5 Shock, drop.- The switchboard shall be operable after the test of 4.6.4. Any physical damage shall be minor only.

3.9 Electrical requirements for Adapter, Connector U-184( )/GT.-

3.9.1 Dielectric strength (See 4.7.1).- The insulation between each terminal and the case of the Adapter, Connector U-184( )/GT shall show no evidence of momentary or intermittent arcing or other indication of breakdown, and there shall be no visible evidence of damage, after application of 400 volts D.C. or peak 60 cycle A.C.

3.9.2 Insulation resistance.- The insulation resistance between the terminals of Adapter, Connector U-184( )/GT shall be not less than 1 megohm. (See 4.7.2)

3.9.3 Visual signal operation.- The glow lamp provided in the Adapter, Connector U-184( )/GT shall flash continuously as long as the alternating current is applied to the terminals. (See 4.7.3)

3.10 Final operation.- The switchboard shall meet the operational inspection specified in 4.8.

3.11 Technical literature, and running spare parts.- Technical literature, and running spare parts shall be furnished as specified by the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2)

3.12 Workmanship.- Workmanship shall be such that the finished product shall meet all the requirements of this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.-

(1) Quality conformance inspection of equipment before preparation for delivery (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.11)

4.3 Preproduction inspection.- This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table I, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, and group B (see tables II and paragraph 4.5.3, respectively).

Table I - Preproduction Inspection

Inspection (For additional preproduction inspection see 4.3)	Req. Para	Insp. Para
Temperature	3.8.2	4.6.1
Relative humidity	3.8.3	4.6.2
Bounce	3.8.4	4.6.3
Shock, drop	3.8.5	4.6.4

4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery:

Item	Where required
Finish	3.4
Marking	3.5

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4.5 Quality conformance inspection of equipment before preparation for delivery.- The contractor shall perform the inspection specified in 4.4 and 4.5.2 through 4.5.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition the Government--at its discretion--may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.4) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Rejected lots.- If an inspection lot is rejected, the supplier may withdraw the lot from further inspection. The supplier may also rework a rejected lot to correct the defects or screen out the defective units and reinspect the lot using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.5.2 Group A inspection.- This inspection, including sampling, shall conform to table II and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operation test shall be performed prior to packaging.

Table II - Group A Inspection

Inspection	Req. Para.	Insp. Para.	AQL Major	AQL Minor
Visual and Mechanical	3.12	4.10	1.0%	4.0%
Electrical				
Dielectric strength	3.9.1	4.7.1		
Insulation resistance	3.9.2	4.7.2	1.0%	*
Visual signal operation	3.9.3	4.7.3		
Final operation	3.10	4.8	0.1%	*

* All Electrical and Operational defects are considered major.

4.5.3 Group B inspection.- This inspection shall consist of the inspection specified for interchangeability in 4.9 and shall conform to the special procedures for small-sample inspection of Standard MIL-STD-105. The AQL shall be 6.5 percent defective and the inspection level shall be L-7 for normal and tightened inspection and L-5 for reduced inspection. The reduced inspection procedure shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met group A inspection.

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4.6 Service conditions tests.-

4.6.1 Temperature.- (See 3.8.2) Subject the Switchboard to the temperature cycle shown on Standard MIL-STD-169. The test of 4.7 shall be made at step 4 and step 8. The equipment shall meet full specification requirements.

4.6.2 Moisture resistance (See 3.8.3).-

4.6.2.1 Test conditions.-

(a) Do not move switchboard from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the switchboard. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

4.6.2.2 Test method.-

(a) Dry at 130° ± 5°F for 24 hours.

(b) Condition at 77° ± 5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.6.2.3 and readjust or realign as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.6.2.3 shall be made during the periods shown on the drawing and shall comply with 4.6.2.3.

(e) After cycling has been completed, condition the switchboard for 24 hours at 77° ± 5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the switchboard. No repair or replacement of parts shall be made. After adjustment, the switchboard shall meet full specification requirements for those measurements specified in 4.6.2.3.

4.6.2.3 Performance.- The insulation resistance measured at a potential of 500 volts d.c. shall not be less than 100,000 ohms between plug terminals and body of plug grounded to water.

4.6.2.4 Failure.- If any switchboard fails to meet the performance specified in 4.6.2.3 during cycling, it shall be realigned or readjusted once. If the switchboard then fails to meet 4.6.2.3 or fails subsequently during cycling, it does not pass the test. In addition, if the switchboard fails to meet full specification requirements as specified in 4.6.2.2(e), it does not pass the test.

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4.6.3 Bounce test (See 3.8.4).- The switchboard shall be tested on the package tester, Type 1000 SC as made by the L.A.B. Corporation, Skaneateles, New York, or equal, as follows:

(a) Cover the tester bed with a panel of 1/2 inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

(b) Place the equipment in its transit case, prepared for field transport, on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

(c) Operate the package tester, shafts in phase, for a total of 3 hours at 284 ±2rpm. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

4.6.4 Drop test.- Drop the switchboard in its transit case a distance of 48 inches on each face, edge and corner for a total of 26 drops. Drops may be made from a quick-release hook, or pendulum tester as made by the L.A.B. Corporation, Skaneateles, New York. The floor or barrier receiving the impact shall be of 2 inch fir backed by concrete or a rigid steel frame. At the conclusion of the test the switchboard shall be operable and meet the requirement of 3.8.5.

4.7 Electrical tests.-

4.7.1 Dielectric strength test.- (See 3.9.1) Subject the Adapter, Connector U-184( )/GT to a potential of 400 volts D.C. or peak 60 cycle A.C., for 10 seconds. Apply the potential between terminals and then between the terminals connected together, and the body of the plug. There shall be no visible evidence of breakdown or damage.

4.7.2 Insulation resistance test.- Immediately following the completion of the dielectric strength test of 4.7.1, test Adapter, Connector U-184( )/GT to determine that the insulation resistance is less than 1 megohm as required in 3.9.2, using a voltohmmeter which operates from a battery supply of not more than 22-1/2 volts.

4.7.3 Visual signal operation test.- Apply 95 volts r.m.s., 17-20 cycle alternating current to the terminals of a completed Adapter, Connector U-184( )/GT through a non-inductive resistance of 10,000 ohms to determine that the glow lamp provided in the adapter flashes continuously as required in 3.9.3.

4.8 Operational inspection.- Immediately prior to packaging of Switchboard, Telephone Manual SB-993( )/GT an operational check shall be made using line-to-line connections to determine that the equipment functions.

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4.9 Interchangeability.- Measure or gage the dimensions listed in the referenced drawings (See 3.3) which effect interchangeability to determine compliance with the physical interchangeability requirement of 3.7. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

4.9.1 Mounting holes.- The mounting holes of each plug holder shall be inspected for accuracy of spacing with respect to the distance between the prongs of Adapter, Connector U-184( )/GT and for accuracy of size with respect to the dimensions of the prongs.

4.9.2 Adapter plugs.- Insert the adapter plugs in the holes of the plug holder and turn the plug holder over so that the mounting surface is held downward and parallel with the plane of the ground The adapter plugs shall not have undue freedom of movement nor fall out of the mounting holes.

4.9.3 Plug holder.- Remove the 7 adapter plugs from the mounting holes and place inside of the plug holder. Shake the plug holder; the adapter plugs shall not rattle nor shall the cover of the plug holder be forced open.

4.10 Visual and mechanical inspection.- The switchboard shall be examined for the defects listed in Standard MIL-STD-252.

4.11 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.-

5.1.1 Level A.- Switchboard, Telephone, Manual SB-993( )/GT shall be preserved, packaged and tested in accordance with the applicable provisions of Specification MIL-P-116 and in a manner that will afford adequate protection against, deterioration and damage during worldwide shipment, handling and open storage.

5.1.2 Level C.- Switchboard, Telephone, Manual SB-993( )/GT shall be afforded preservation and packaging in accordance with the suppliers' normal commercial practice.

5.2 Packing.- Switchboard, Telephone, Manual SB-993( )/GT packaged as specified, shall be packed in shipping containers conforming to the requirements of the specifications referenced below for the designated level. The gross weight shall not exceed the weight shown unless an individually packed item exceeds that amount. Closure and strapping shall be as prescribed in the applicable box specification or appendix thereto, except that bands of reinforced, pressure-sensitive adhesive tape, 1/2 inch wide and conforming to Specification PPP-T-97 for Grade IV, shall be used for fiberboard boxes in lieu of metal strapping.

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5.2.1 Level A.-

BOX SPECIFICATION		MAX. GROSS WT.
Fiberboard (Type I, Class 2)	PPP-B-636	Table VI
Nailed Wood (Style 4, Class 2)	PPP-B-621	200 lbs.
Wirebound (Style 2 or 3, Class 2)	PPP-B-585	200 lbs.
Wood Cleated Fiberboard (Overseas)	PPP-B-591	200 lbs.
Wood Cleated Plywood (Overseas)	PPP-B-601	200 lbs.
Wood Cleated, Veneer, Paper Overlaid (Overseas)	MIL-B-10377	200 lbs.

5.2.2 Level B.-

BOX SPECIFICATION		MAX. GROSS WT.
Fiberboard (Type I, Class I)	PPP-B-636	Table I
Nailed Wood (Style 4, Class I)	PPP-B-621	200 lbs.
Wirebound (Style 3, Class I)	PPP-B-585	200 lbs.
Wood Cleated Fiberboard (Domestic)	PPP-B-591	200 lbs.
Wood Cleated Plywood (Domestic)	PPP-B-601	200 lbs.
Wood Cleated, Veneer, Paper Overlaid (Overseas)	MIL-B-10377	200 lbs.

5.2.3 Level C.- Switchboard, Telephone Manual SB-993( )/GT shall be packed in shipping containers of the type, size and kind commonly used for the purpose in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the rules or regulations of the common carrier, applicable to the mode of transportation.

5.2.4 Pilot pack.- When Level "A" packaging and packing is specified above, one acceptable model of Switchboard, Telephone Manual, SB-993( )/GT will be pilot packed in a manner that will pass the preproduction tests prescribed in Specification MIL-P-116. Illustrations of the packaging and packing procedure, together with Bills of Material, will be prepared and furnished in accordance with the provisions of Signal Corps Drawing SC-A-362100.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of Standard MIL-STD-129.

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6. NOTES

6.1 Intended use.- Switchboard, Telephone, Manual SB-993( )/GT is intended to provide connections among 6 telephone lines employing field wire, to provide visual signals when ringing current is applied to such lines, and for use as an emergency substitute for a 6 line magneto telephone switchboard.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing requirement for shipment. (Level A, level B or level C.)

(d) The specific paragraphs of Section 5 which are applicable to the particular procurement.

(e) Preproduction pack(s) as follows: (

Makeup of pack(s).
Number of each kind of pack to be submitted.
Inspection to be performed thereon.

(f) Marking and shipping of samples.

(g) Place of final inspection.

(h) Technical literature required. (See 3.11)

(i) Quantity of running spare parts required. (See 3.11)

(j) The statement of treatment referenced in 3.6.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: SB-993W/GT. As soon as possible after the award of the contract, the contractor should apply to the Government office specified in the contract for such information. (See 1.1).

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6.4 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.5 International standardization agreements.- Provisions of this specification are the subject of international standardization agreements, ABC Number 62. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Office so that appropriate action may be taken respecting the international agreements concerned.

Project Number 5805-A043

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